

GOVERNMENT ART'S AND SCIENCE COLLEGE - Alangudi

College code - BDU752

Department name - Computer science

Project name - Script controlled ACL

submitted by,

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SCRIPT - CONTROLLED ACL

phase : Documentation phase

Description :

This lab demonstrates how to implement a script-controlled Access Control List (ACL) to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the EEE branch are allowed to view records where the Branch field is set to EEE, while administrators retain full access.

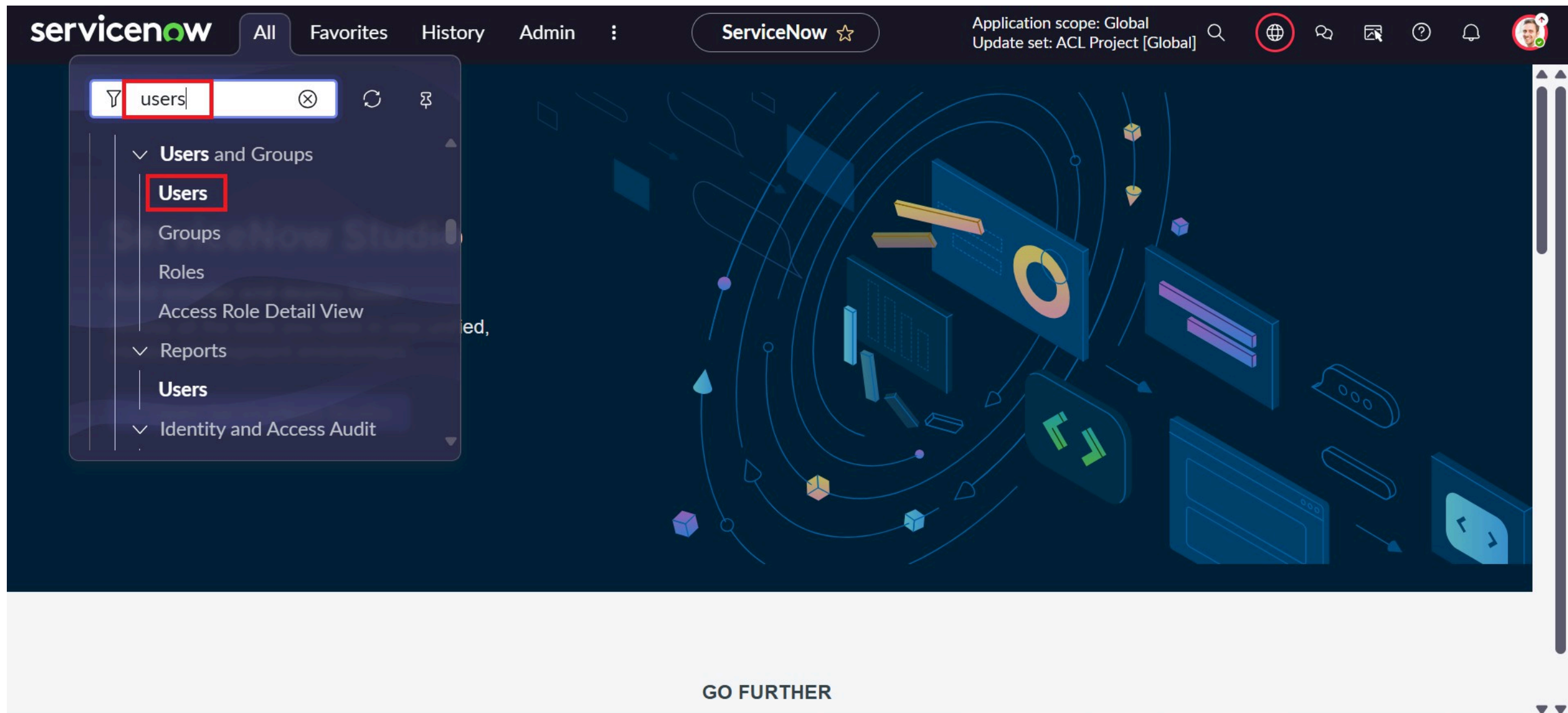
Timer :

3 hrs

Milestone-1 Creation of Users and Roles

User Creation:

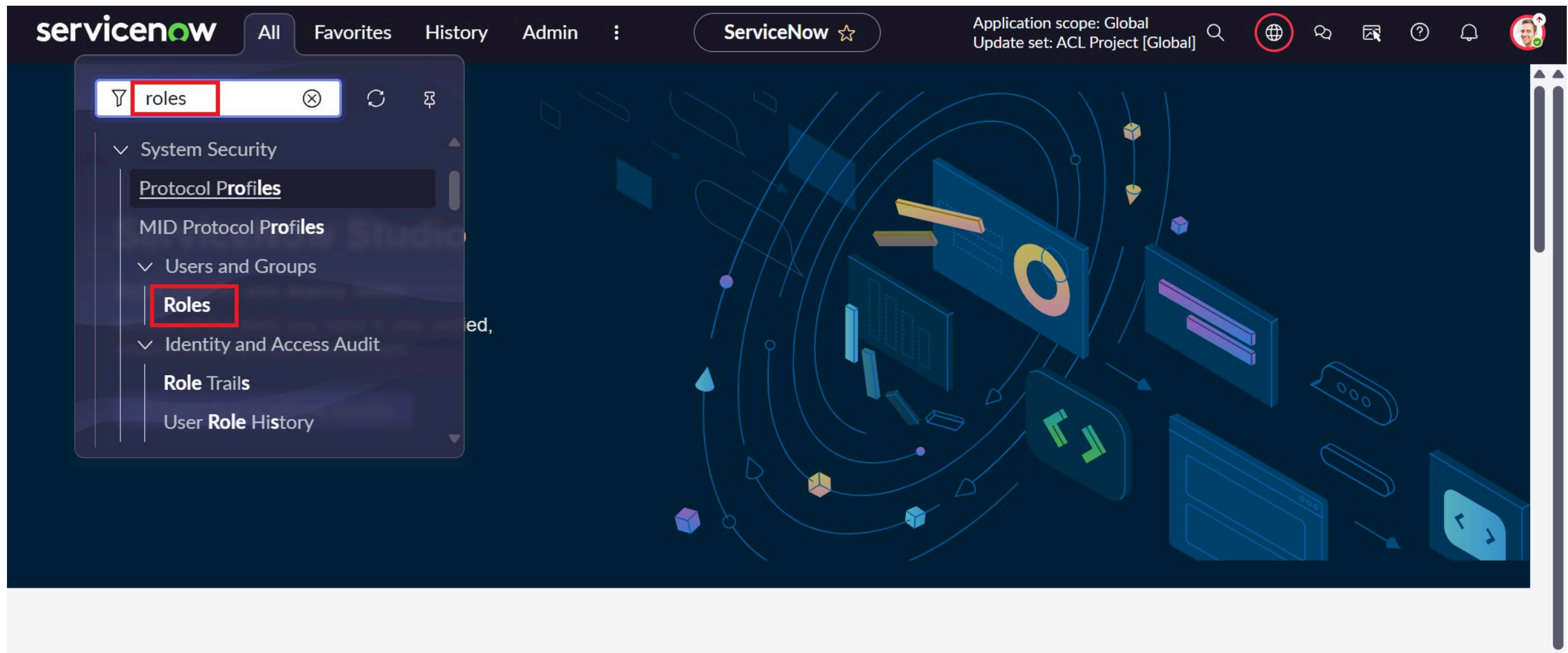
1. Log in to the ServiceNow instance with admin access.
2. Create a test user:
 - Navigate to User Administration → Users
 - Click New
 - User ID: EEE User
 - First name: EEE
 - Last name: User
 - Email: eeeuser@gmail.com
 - Save the record



The screenshot shows the 'User Administration' form for a user named 'User: EEE User'. The form is divided into two main sections. The left section contains fields for 'User ID' (EEEUser), 'First name' (EEE), 'Last name' (User), 'Title', 'Department', and 'Location'. The right section contains fields for 'Email' (eeeuser@gmail.com), 'Identity type' (Human), 'Language' (-- None --), 'Calendar integration' (Outlook), 'Time zone' (System (Asia/Kolkata)), 'Date format' (System (yyyy-MM-dd)), 'Business phone', and 'Mobile phone'. At the bottom, there are checkboxes for 'Password needs reset', 'Locked out', 'Active' (checked), and 'Internal Integration User'. The 'Update' button at the bottom left is highlighted with a red box.

Role Creation

- Navigate to User Administration → Roles
- Click New
- Name: Enter role name as bb1
- Description: Brief role purpose
- Click Submit
- Create other roles also like bb2,bb3,bb4.(follow the same procedure)
- Assign all the roles to the EEE User



< ≡ Role BB1 Update Delete ↑

Name Application ?

Elevated privilege ☐

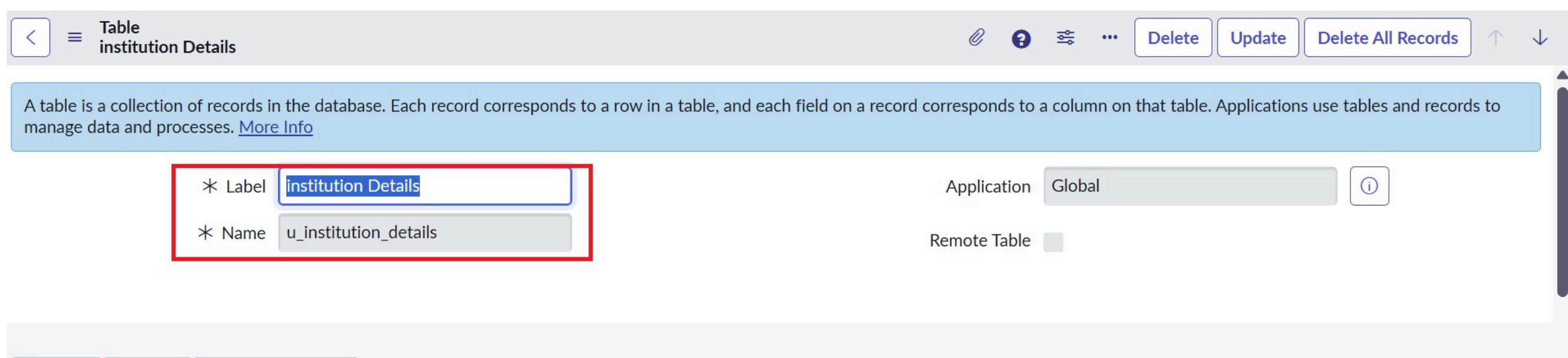
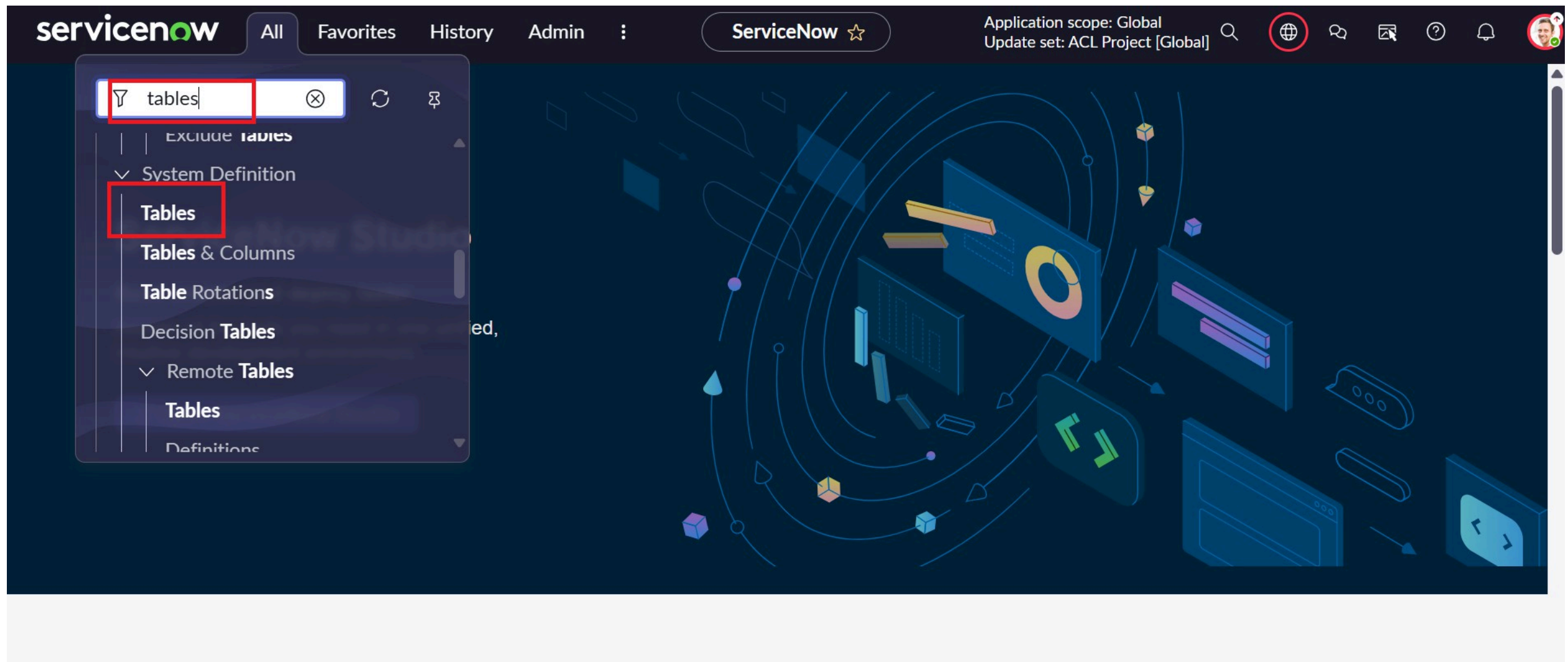
Description

Update Delete

[SN Utils](#) [Versions \(1\)](#)

Tables Creation

1. In the Application Navigator, search for System Definition → Tables.
2. Click New and create a custom table with the following details:
 - Label: Institution Details
 - Name: u_institution_details
 - Extends for : false



3. Click Submit.
4. Add the following fields to the u_institution_details table:
 - Student Roll Number– Auto Number
 - Student Name – Reference- User
 - Faculty Name – Reference- User
 - Branch – Choice (ECE, EEE, CSE)
 - Email– String
 - Phone Number– String
 - Description– Multi String
5. Save the Table

Table Columns for text Search					1 to 13 of 13
Dictionary Entries					
	Column label	Type	Reference	Max length	Default value
	Updates	Integer	(empty)	40	
	Created by	String	(empty)	40	
×	Student Roll Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();
×	① Description	String	(empty)	1,000	
	Updated by	String	(empty)	40	
×	Phone Number	String	(empty)	40	
	Sys ID	Sys ID (GUID)	(empty)	32	
	Created	Date/Time	(empty)	40	
×	Faculty name	Reference	User	32	
×	Branch	String	(empty)	40	
	Updated	Date/Time	(empty)	40	
×	Student Name	Reference	User	32	
×	Email	String	(empty)	40	
+	Insert a new row...				

- Create multiple records in the Student Records table with different branch values (ECE, EEE, CSE).

institution Details

Student Roll Numbe

Search

Actions on selected rows...

All

Student Roll Number	Faculty name	Student Name	Branch	Email	Phone Number
Search	Search	Search	Search	Search	Search
INS0001006	ITIL User	Praneeth	ECE	praneeth@gmail.com	8989090906
INS0001008	Malla Sravan kumar	Abraham Lincoln	CSE	sravan@gmail.com	8978908989
INS0001005	ITIL User	Srinu Nandipalli	ECE	srinu@gmail.com	9089909090
INS0001007	Abel Tuter	Guna Sai	EEE	saiguna@gmail.com	9090898765
INS0001001	Abraham Lincoln	Yaswanth Nikku	CSE	yaswanth@gmail.com	8989898989
INS0001004	Abel Tuter	Prashanth Reddy	EEE	prashanth@gmail.com	9090909090

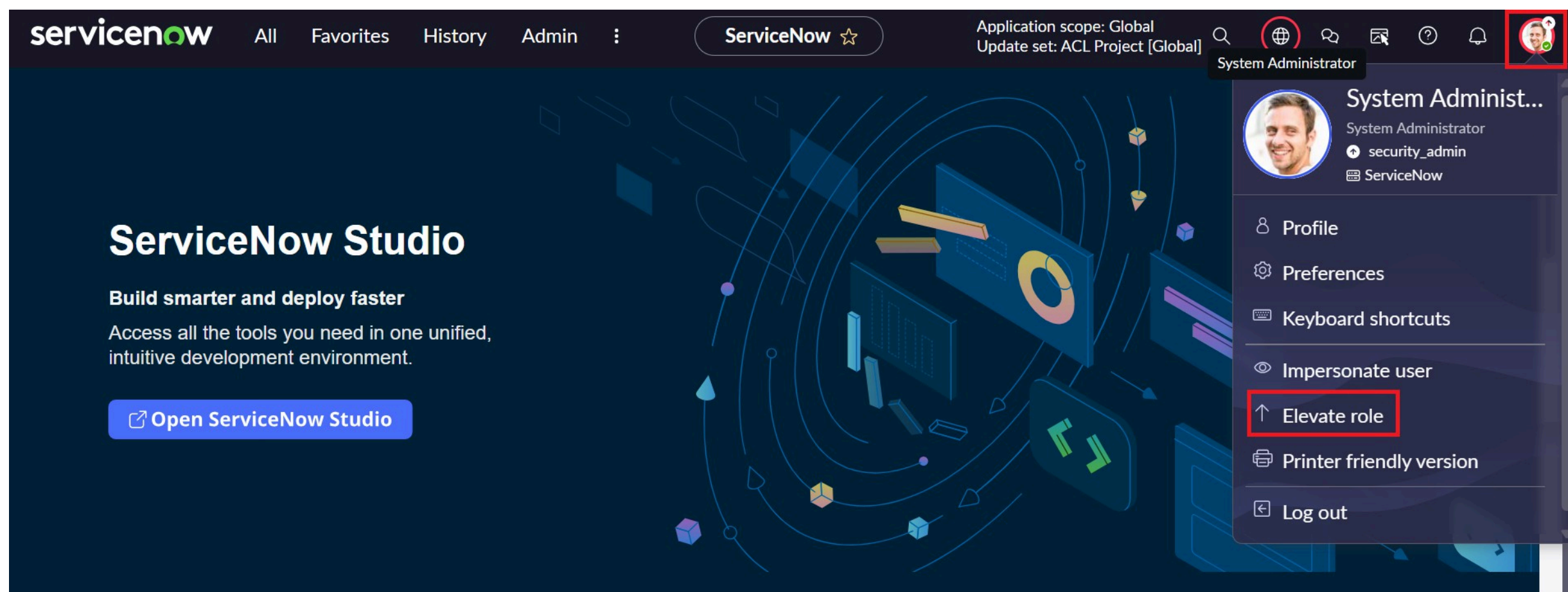
Creation of Access Control Lists(Read ACL)

- In the Application Navigator, search for System Security → Access Control (ACL).
- Elevate “Security_admin” role
- Click New to create a new ACL.
- Enter the following details:
 - Type: record
 - Operation: read
 - Name: u_institution_details
 - Active: true

- 0 Advanced: true
- 5. In the Requires role related list, add:
 - 0 bb1 (custom role)
- 6. Add Data condition as Branch is EEE
- 7. In the Script field, add the following script:

```
(function () {  
  
// Allow admin users full access if  
(gs.hasRole('admin')) {  
    return true;  
}  
  
// Allow only EEE branch users to see EEE records if  
(gs.hasRole('bb1')){  
    return true;  
}  
  
// Deny access for all others  
return false;  
})();
```

- 8. Click Submit to save the ACL.



Elevate role

Close dialog

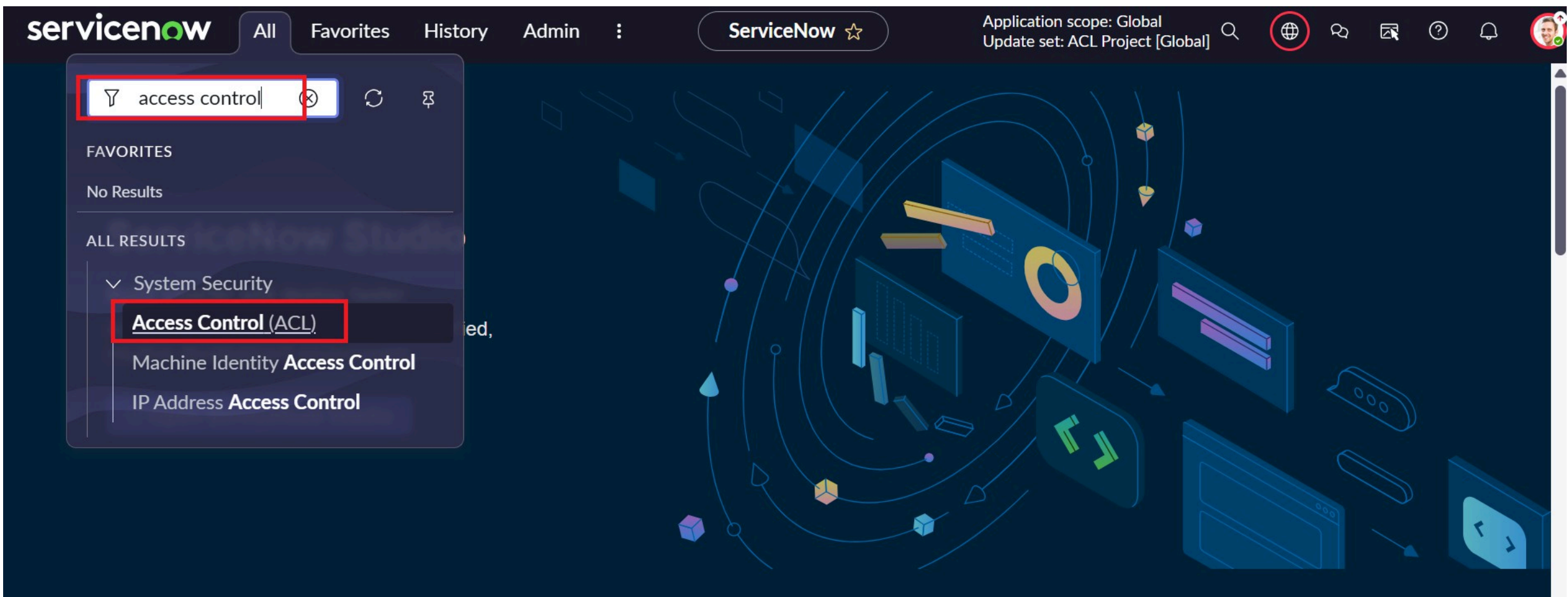
Elevate a role by adding privileges, which end when you log out. [Learn more](#)

AVAILABLE ROLES

☒ security_admin

Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel Update



Access Controls

Updated

Search

Actions on selected rows...

New

All

☐		Name	Decision Type	Operation	Type	Active	Updated by	Updated
☐								
		sn_try_build_agent_message	Deny Unless	read	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_freemium_usage_log	Deny Unless	read	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_conversation	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_conversation	Deny Unless	write	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_message	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_event_telemetry	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_conversation	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_task_telemetry	Deny Unless	write	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_task_telemetry	Deny Unless	read	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_event_telemetry	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_task_telemetry	Allow If	read	record	true	system	about 3 months ago
☐		sn_try_build_agent_knowledge_source	Allow If	write	record	true	system	19-10-2025 22:23:04
		sn_try_build_agent_event_telemetry	Allow If	report_view	record	true	system	19-10-2025 22:23:04

61

to 80 of 32,702

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Access Control

u_institution_details

Update

Delete

* Type

record

* Operation

read

Decision Type

Allow If

Admin overrides

☒

Protection policy

-- None --

Application

Global

Active

☒

Advanced

☒

* Name

institution Details [u_institution_details]

-- None --

Description

Applies To

No. of records matching the condition: 6

Add Filter Condition

Add OR Clause

-- choose field --

-- oper --

-- value --

Condition

No. of records matching the condition: 2

Add Filter Condition

Add OR Clause

Branch

is

EEE

AND

OR

Advanced Condition

Controlled by References

Script

Turn on ECMAScript 2021 (ES12) mode

```
1 (function() {
2   if (gs.hasRole('bb1')) {
3     return true;
4   } else {
5     return false;
6   }
7 }());
```

- 9. Impersonate EEE users and open the Student Records list.
- 10. Copy the table name u_institution_details.list/LIST and paste in the All menu.
- 11. Impersonate Admin and open the Student Records list.

Verify :

- User with bb1 role can view only EEE branch records

institution Details

Student Roll Numbe

Search

Actions on selected rows...

All

☐	Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
	INS0001007	EEE		saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
	INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

- User without the role cannot view any records

institution Details

Student Roll Numbe

Search

Q

Student Roll Number ▲

Branch

Description

Email

Faculty name

Phone Number

Student Name

Search

Search

Search

Search

Search

Search

Search

?

No records to display

- Admin user can view all records regardless of branch

institution Details

Student Roll Numbe

Search

Actions on selected rows...

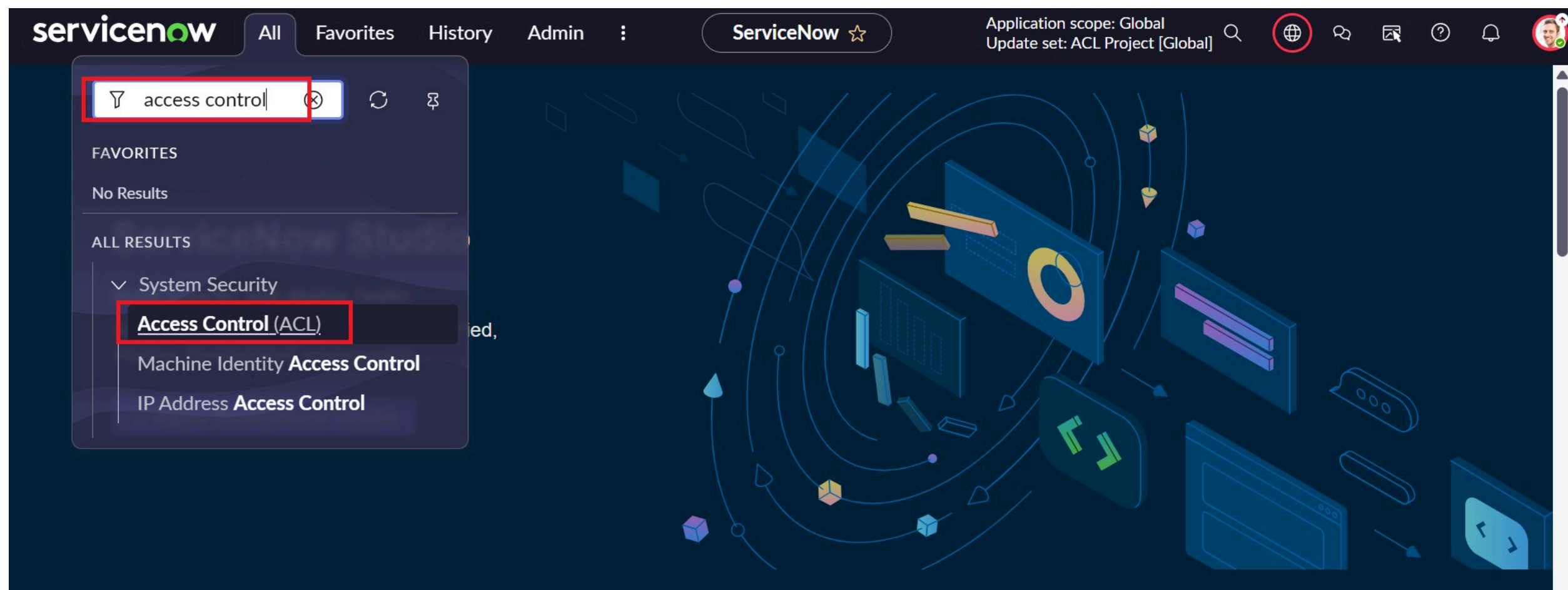
New

All

Student Roll Number	Faculty name	Student Name	Branch	Email	Phone Number
Search	Search	Search	Search	Search	Search
INS0001006	ITIL User	Praneeth	ECE	praneeth@gmail.com	8989090906
INS0001008	Malla Sravan kumar	Abraham Lincoln	CSE	sravan@gmail.com	8978908989
INS0001005	ITIL User	Srinu Nandipalli	ECE	srinu@gmail.com	9089909090
INS0001007	Abel Tuter	Guna Sai	EEE	saiguna@gmail.com	9090898765
INS0001001	Abraham Lincoln	Yaswanth Nikku	CSE	yaswanth@gmail.com	8989898989
INS0001004	Abel Tuter	Prashanth Reddy	EEE	prashanth@gmail.com	9090909090

Creation of Access Control Lists(Create ACL)

1. Click New to create a new ACL(Create)
2. Enter the following details:
 - Type: record
 - Operation: create
 - Name: u_institution_details
 - Active: true
3. In the Requires role related list, add:
 - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



Access Controls

Updated

Search

Actions on selected rows...

New

All

Search

Search

Search

Search

Search

Search

Search

sn_try_build_agent_message

Deny Unless

read

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_freemium_usage_log

Deny Unless

read

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_conversation

Deny Unless

delete

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_conversation

Deny Unless

write

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_message

Deny Unless

delete

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_event_telemetry

Deny Unless

report_view

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_conversation

Deny Unless

report_view

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_task_telemetry

Deny Unless

write

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_task_telemetry

Deny Unless

read

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_event_telemetry

Deny Unless

delete

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_task_telemetry

Allow If

read

record

true

system

about 3 months ago

sn_try_build_agent_knowledge_source

Allow If

write

record

true

system

19-10-2025 22:23:04

sn_try_build_agent_event_telemetry

Allow If

report_view

record

true

system

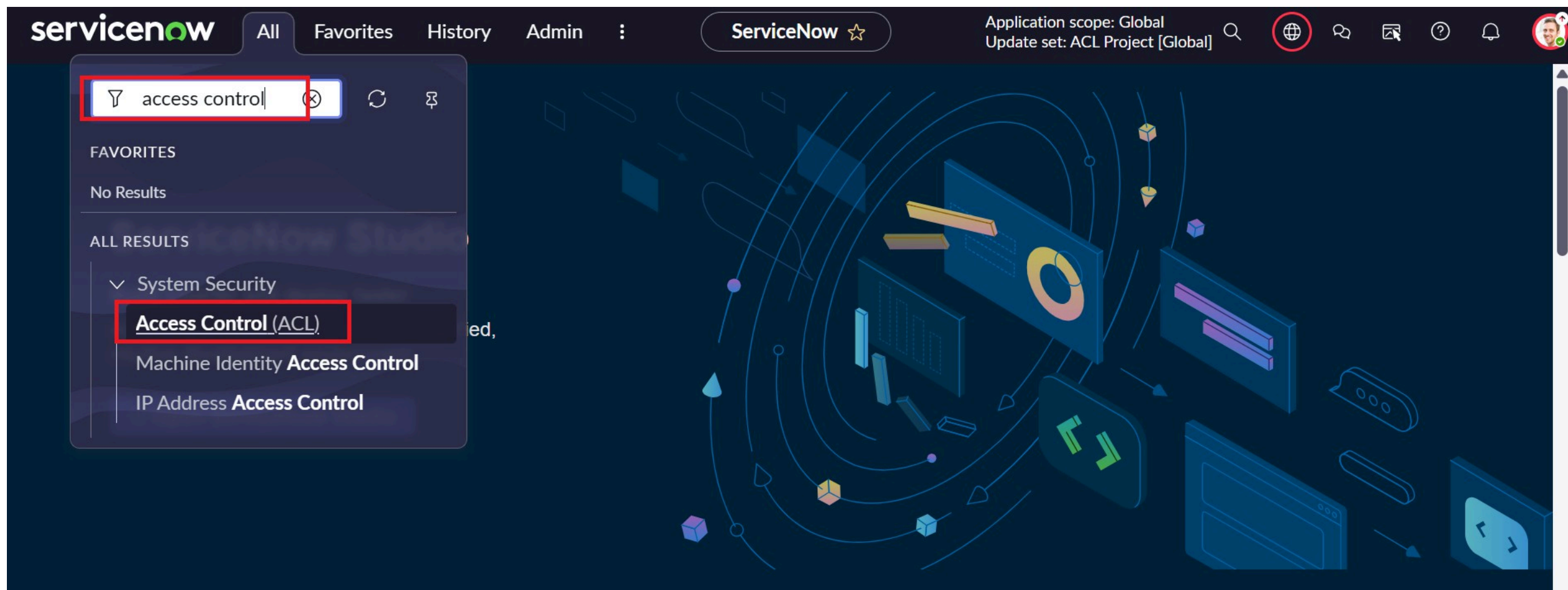
19-10-2025 22:23:04

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0 Name: u_institution_details

- 0 Active: true
14. In the Requires role related list, add:
 - 0 bb3 (custom role)
15. No need to add any Data condition
16. Save the ACL Record



Access Controls

Updated

Search

Actions on selected rows...

New

All

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role
Role
× BB3
+ Insert a new row...

- User with bb1 &bb2 &bb3 roles can view and edit the EEE branch records along with New Button

≡

🔍

💬

institution Details

Student Roll Numbe ▾

Search

⚙️

📄

Actions on selected rows... ▾

New

All

☐	🔍	Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
		INS0001007	EEE		com	Abel Tuter	9090898765	Guna Sai
		INS0001004	EEE		il.com	Abel Tuter	9090909090	Prashanth Reddy

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

☐		Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
		INS0001007	EEE	ACL is Executed	saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
		INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

Creation of Access Control Lists(Delete ACL)

17. Click New to create a new ACL(Delete)

18. Enter the following details:

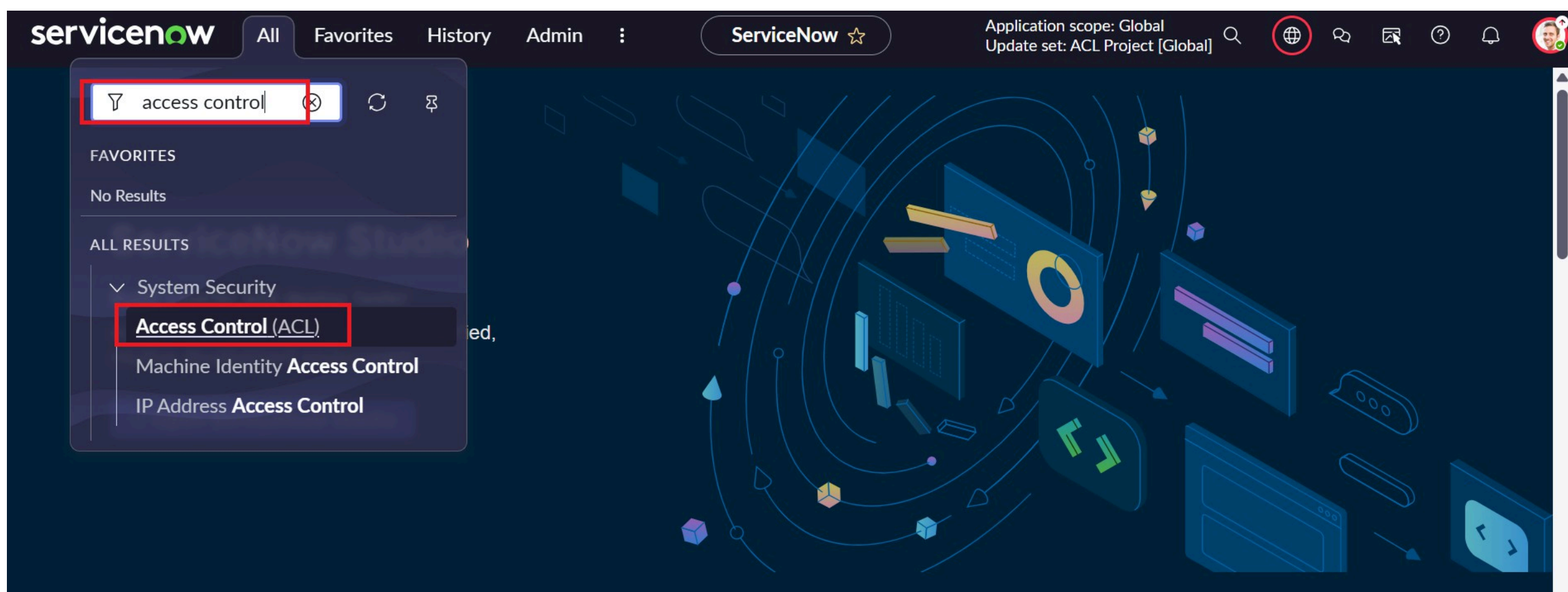
- 0 Type: record
- 0 Operation: Delete
- 0 Name: u_institution_details
- 0 Active: true

19. In the Requires role related list, add:

- 0 bb4 (custom role)

20. No need to add any Data condition

21. Save the ACL Record



Access Controls

Updated

Search

Actions on selected rows...

New

All

	Name	Decision Type	Operation	Type	Active	Updated by	Updated
	Search	Search	Search	Search	Search	Search	Search
	sn_try_build_agent_message	Deny Unless	read	record	true	system	19-10-2025 22:23:04
	sn_try_build_agent_freemium_usage_log	Deny Unless	read	record	true	system	19-10-2025 22:23:04
	sn_try_build_agent_conversation	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
	sn_try_build_agent_conversation	Deny Unless	write	record	true	system	19-10-2025 22:23:04
	sn_try_build_agent_message	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
	sn_try_build_agent_event_telemetry	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04
	sn_try_build_agent_conversation	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04
	sn_try_build_agent_task_telemetry	Deny Unless	write	record	true	system	19-10-2025 22:23:04
	sn_try_build_agent_task_telemetry	Deny Unless	read	record	true	system	19-10-2025 22:23:04
	sn_try_build_agent_event_telemetry	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
	sn_try_build_agent_task_telemetry	Allow If	read	record	true	system	about 3 months ago
☐	sn_try_build_agent_knowledge_source	Allow If	write	record	true	system	19-10-2025 22:23:04
	sn_try_build_agent_event_telemetry	Allow If	report_view	record	true	system	19-10-2025 22:23:04

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Access Control

New record

Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type

record

* Operation

delete

Decision Type

Allow If

Admin overrides

☒

Protection policy

-- None --

* Name

institution Details [u_institution_details]

-- None --

Description

Applies To

No. of records matching the condition: 6

Add Filter Condition

Add OR Clause

-- choose field --

-- oper --

-- value --

Application

Global

Active

☒

Advanced

☐

Requires role

Role

BB4

Insert a new row...

Security Attribute Condition

Verify :

- Users with bb1 &bb2 &bb3&bb4 roles can see, create and edit along with delete the
EEE branch records.

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

☐	Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
	INS0001007	EEE	ACL is Executed	saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
	INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

Student Roll Number

Branch

Description

Email

Faculty name

Phone Number

INS0001007

EEE

ACL is Executed

saiguna@gmail.com

Abel Tuter

90908987

INS0001004

EEE

ACL is Executed

prashanth@gmail.com

Abel Tuter

90909090

Actions on selected rows...

Delete

Delete with preview...

Create Application File

Assign Tag:

New tag

Now Platform Security

Strategic Portfolio Management

Field Service Management

Now Platform

Security

IT Asset Management

IT Service Management

IT Operations Management

More...

institution Details

Student Roll Numbe

Search

Actions on selected rows...

All

☐	Student Roll Number	Branch	Description	Faculty name	Phone Number	Student Name
☒	INS0001007	EEE	ACL is Executed	Abel Tuter	9090898765	Guna Sai
☐	INS0001004	EEE	ACL is Executed	Abel Tuter	9090909090	Prashanth Reddy

Confirmation

Delete the selected record(s)?

Cancel

Delete

Close

Outcome :

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal

interfaces.

Github link :

<https://github.com/ahamedibrahim07989-cell/Script-Controlled-ACL-Restrict-Record-Access-Based-on-Field-Value.git>